

Enhancing the Accuracy of Forest Monitoring Through AI to Reduce Carbon Footprint

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Abstract— Global Forests acquires almost 30% of total land surface and it holds huge value for the betterment of environment and humanity. In recent years, shocking image of forest destruction has been observed in various regions such as “Australia”, “western United States”, “Europe”, and the “Amazon rainforest”, which have engulfed in fire. As a result, it has become the first propriety for people and responsible authority to manage and maintain forest across the world to save humanity. In this process, proper mapping and monitoring of global forests are an extremely important factor. As a result, intervention of Artificial Intelligence (AI), in order to identify and monitor forests with huge conservation value along with, huge carbon stock is necessary for the entire environment. The main aim of this research was to identify the role of AI to enhance the accuracy of forest monitoring process for reducing carbon footprint. The literature review section of this research paper have focused on different tools and techniques based on AI technology, used for Forest Monitoring process and the way under which it reduces overall carbon footprint. This research paper has been conducted through a mixed method approach, where an interview and secondary sources both have been used to meet the research objectives. The interview process has been developed by engaging three employees of forest monitoring agencies in India to identify their viewpoint regarding the effect of AI in forest monitoring process. As per the findings from interview and secondary information, AI plays a huge role in monitoring forest degradation process. However, some technical deficiencies are there, that can be mitigated with continuous development strategies.

Keywords— Forest Degradation, Forest Monitoring, Artificial Intelligence, Carbon footprint, Carbon emission, Remote sensing,

Machine Learning, Wireless Network, Interactive Web-based Near Real-Time Forest Monitoring System, Forest Fire Detection

I. INTRODUCTION

Background

Deforestation has been a major issue within the environmental aspect and illegal logging in the rainforest is now considered as the traditional organized crime. As per data, almost 31% of the plant is covered with forest and it is home to almost 80% of species of animals and plants across the world. Forests also provide elements for the livelihood of over 1.6 million people. However, still the rate of deforestation is increasing day-by-day and this is destroying the rich biodiversity of these 1.6 million people and species of animals and plants across several parts of this world. In terms of Amazon forest, the range of forest destruction is shocking and this was primarily driven by illegal activities including “land invasions”, “induced forest fires”, which was driven in order to clear land for “agriculture”, “ranching” as well as “land speculation” [1].

Illegal mining and logging are another reason behind this forest destruction. Deforestation created a major impact on the environment and accounts for almost 17% of global carbon emission. This global crisis not only accounts for a huge environmental loss, it also develops a huge economic loss [1].

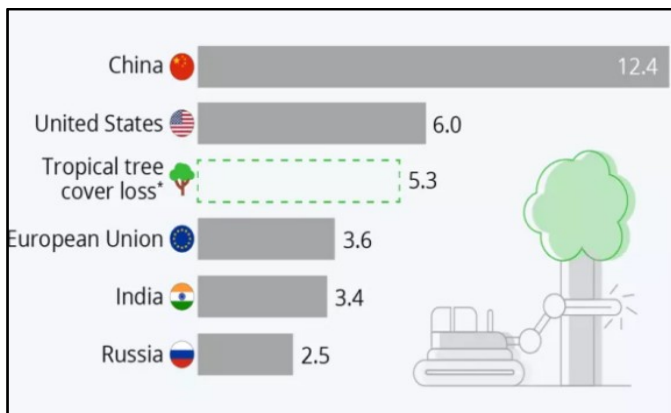


Figure 1: Carbon Emission by Deforestation in Different Countries
 (Source: [2])

As per the above report, it has been observed that China is in the first position, having emitted 12.4 Gt of carbon per year due to deforestation. India is in the 4th position with 3.4 Gt per year. Lack of adequate forest monitoring is a major reason behind this crisis. However, in recent years, promising innovations are introduced for forest monitoring through Artificial Intelligence (AI) such as deep learning, machine learning and many others [1]. This research will focus on identifying and describing use of these AI tools in the forest monitoring process.

Aim

Aim of this research is to identify the role of AI to enhance the accuracy of forest monitoring process for reducing carbon footprint

Objectives

- To identify use of different tools and techniques within forest monitoring process
- To examine the effectiveness of using these tools along with some challenges
- To recommend best strategic applications for enhancing effectiveness of forest monitoring process

II. LITERATURE REVIEW

Use of “Interactive Web-based Near Real-Time Forest Monitoring System”

Tropical Forest plays a major role within the stabilization of climate, providing food, water, wood to large numbers of animal and plant species, along with humans. It also provides habitats for the purpose of biodiversity. Deforestation along with forest degradation are widely acknowledged as a major environmental issue by the scientific community across the world, which is responsible for a major increase of greenhouse gasses in the atmosphere [3]. This develops a major change within the global hydrological cycle. In recent times, most of the forecast monitoring system focuses on activity data, which is based on “forest cover changes” and in these two approaches are used such as Top down and bottom-up approach [3].

Top-down approach mainly focuses on change in satellite system and bottom-up approach focuses on utilizing frond

observation with the help of government agencies. Satellite data is important for near real-time (NRT) forest monitoring. The NRT system contributes a better result in the process of forest monitoring with AI tools. A web-based “Interactive Forest Monitoring System (IFMS)” has been designed in order to assist local stakeholders for the identification of forest change locations in NRT and help them to gather optimized information regarding changes.

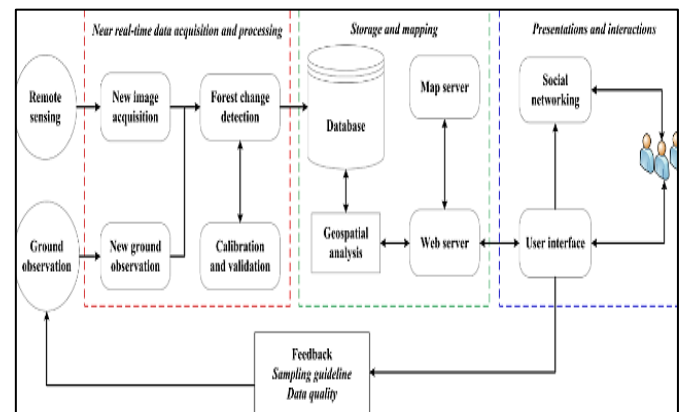


Figure 2: Design Architecture of “Interactive Web-based NRT Forest Monitoring System”
 (Source: [3])

As per the above figure, the entire software design of this multilayer forest monitoring system includes 4 functional modules such as NRT data acquisition and processing”, “storage and mapping”, “presentation and interaction”, and “feedback”. These 4 modules mainly comprise the initial functionality of the server side. Moreover, on the other end, the users such as local stakeholders or any forest professionals send requests to the server through their “mobile” or any “web-based graphic user interface” [3]. This particular server carries a spatial analysis and transfers the result towards the client for getting their feedback and interacting with them with data visualization. This particular framework is scalable and all the 4 modules are established with the help of several AI-based tools such as remote sensing, server recognition, mapping server, and other database related features and analysis process. As per a performance analysis of this framework, it has been observed that accuracy of satellite-based forest changing alerts has increased to 77% during its operational phase, which seems to be effective for preventing deforestation to some extent through forest monitoring process.

Use of Machine-Learning Algorithm for the Susceptibility of Forest Fire

Forest fire can be suspected based on multi-sensor data, as well as advanced computer intelligence, which also includes AI and several machine learning algorithms. All of these tools have improved the accuracy of forest monitoring systems and forest fire prediction. It is always important to detect accurate data regarding the location of fire within a forest, and extent of fire in order to identify the response time according to the intensity [4]. These features are effective to prevent forest fires.

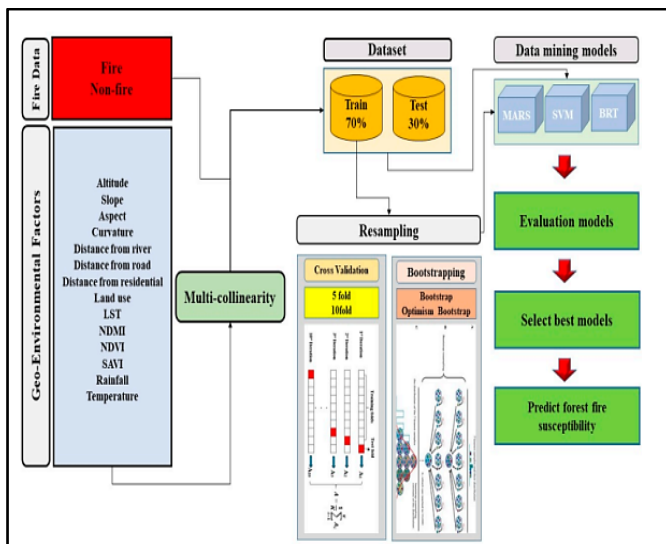


Figure 3: Overall Framework for Proposed Methodology to detect Forest Fire
(Source: [4])

The above flow diagram has shown a proposed methodology to locate fire in a forest. It shows the flow of data processing and analysis steps while predicting forest fires. Firstly, Fire data is collected, and then a “multicollinearity model” is developed based on some “geo-environmental factors”. After that, train and test data are classified. Train data is transferred to the resampling process with cross validation and bootstrapping. After the resampling process, one of these “machine learning models” such as “SVM (support vector machine)”, “BRT (boosted regression tree)”, “MARS (multivariate adaptive regression splines)” are used for data mining [4]. All the models are evaluated, and then the most suitable model is chosen for predicting “Forest Fire Susceptibility”.

Literature Gap

The above literatures have focused on use of machine learning technologies to detect forest fire and use of “NRT Forest Monitoring System” to prevent deforestation with effective accuracy. However, challenges related to these systems are not significantly addressed in this section. As a result, the research paper will gather more data to discuss additional features of AI tools within the forest monitoring process and its benefits.

III. METHODOLOGY

Research Methods

Research philosophy is important for a research paper in order to identify reliable knowledge regarding the research objectives. In this research paper, “Pragmatism Philosophy” has been used to gather data based on human experience and consequences ([5]). Under pragmatism philosophy, this particular research paper will utilize data from human experiences regarding the use of AI tools in the forest monitoring system. Apart from this, a descriptive research design has been used in this study to gather information based on research objectives such as use of AI tools for enhancing the accuracy within forest monitoring process and insights of

different tools used in this process [6]. Inductive research approach allows this research paper to develop knowledge and theories regarding use of AI in the forest monitoring process [6 & 7].

Data Collection and Data analysis

Primary and Secondary both the data collection methods are considered within this research paper for gathering information from different perspectives. Primary data collection method has provided first-hand data with the help of an interview process [8]. The interview has been done on three employees, who work as forest professionals in India and their outlook on the use of AI in forest monitoring has been identified. Moreover, secondary data is collected to explore additional information regarding this aspect with a large amount of exciting secondary sources such as Google Scholar and Websites [9]. These collected data from secondary sources are analyzed after building some themes, as themes are effective to gather hidden patterns and information within the data [10].

IV. RESULT AND DISCUSSION

Primary Data Analysis

Questions	Answers
1) What are the challenges that are creating obligations to prevent forest deforestation?	R1: Lack of time, lack in skilled professional team, lack in technical advances are the main reason behind deforestation, which induce carbon emission R2: Poor government policies are the major reason behind the long-term issues within the forest monitoring process R3: Lack of accuracy within the existing forest monitoring system are the main reason behind wrong interpretation
2) What are the factors that are important according to you to implement AI tools for forest monitoring?	R1: I think, an well-established technical expert team is essential before implementing AI tools within the process of forest monitoring. R2: According to me, effective knowledge regarding statistics and use of machine learning algorithm are the main factor to use AI tool for enhancing the accuracy of Forest Monitoring R3: I think, the main factor that is important to use AI tools within forest monitoring process is skilled technicians, skilled forest professionals and government intervention.

3) Do you think that using AI tools and machine learning techniques will increase the accuracy of forest data and how?	R1: According to me, using AI tools can enhance overall accuracy level within the forest monitoring process with proper data mining. R2: Modern technology and algorithms are now effective to transfer proper alert to the users in exact time. R3: I think, the use of modern AI-based tools within the forest monitoring system are effective, because these tools have the efficiency to manage large manage of complex data to run the statistical calculation
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Table 1: Interview Result

Thematic Analysis

Theme 1: Remote Sensing for Forest Monitoring Process involves different challenges and slow progression in forest restoration

Remote sensing is used as a tool to deal with different real world problems and the forest monitoring process is one of them. Remote monitoring has been used within different applications such as “global forest change”, “global forest resources assessment”, and forest fire detection. Remote sensing with big data is effective to accomplish targets related to forest monitoring, which are not possible with traditional methods.

However, the main challenge within this remote sensing process for forest minting is big data analysis.

The main challenge in data computing is to develop a more heterogeneous system to integrate resources in several locations. In terms of collaboration, data access and connectivity are the main obstacles. On the other hand, in terms of data methodology, it is challenging to deliver right data to right computing nodes. Some other individual big data challenges are also there such as data identification, data transport, data representation and many others [11]. Apart from this, in terms of Forest Monitoring Process, scalability for integrating “multiple disparate management systems” for different satellites for a remote-sensing data centre is a major challenge. These challenges can deliver inaccurate data regarding forest cover changes.

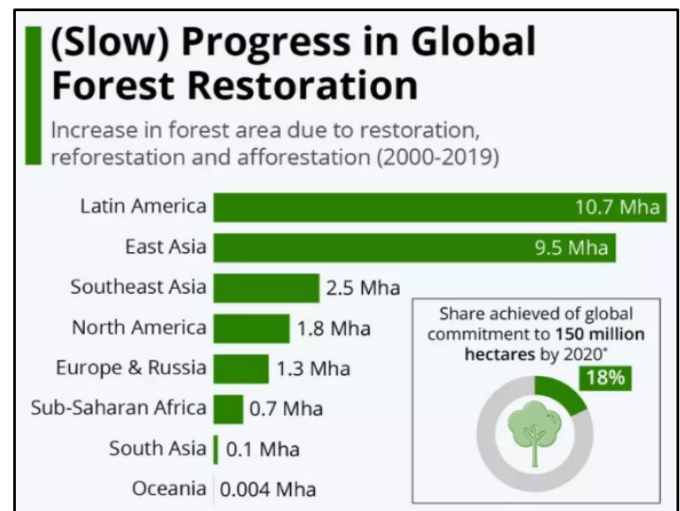


Figure 5: Slow Progress in Forest Restoration across the world
(Source: [12])

As per the above data, South Asian countries including India are in a lower position in terms of forest restoration and a major reason behind this can be poor forest monitoring systems than developed regions such as Latin America and East Asian countries [12]. Technological advancement is needed to accelerate the process and reduce carbon emission.

Theme 2: Wireless Sensor Network detects early Forest Fire

In respect to global warming, the number of forest fires is increasing day-by-day. In order to support stakeholders such as fire soldiers, a model has been established for early detection of forest fires [13]. The system that has been developed for using wireless sensor networks in early detection of forest fire has used machine-learning algorithms.

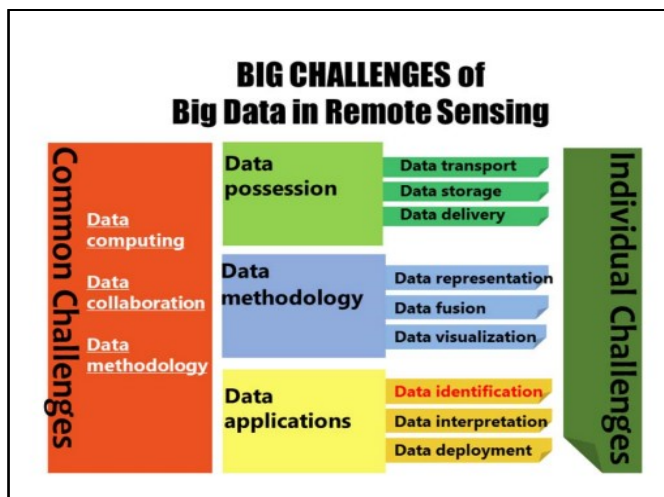


Figure 4: Challenges related to Big Data
(Source: [11])

The above figure shows different challenges related to big data analysis while using remote sensing in any application including forest monitoring. Data computing, data collaboration and data methodology are common challenges.

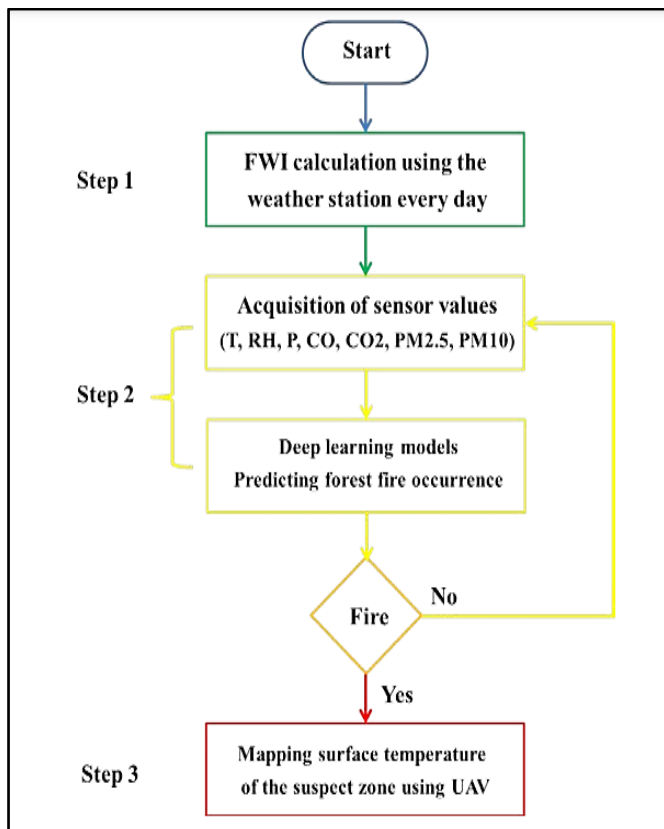


Figure 6: Flowchart for Algorithm used in Forest Fire detection
 (Source: [13])

The above algorithm includes three steps one is “Forest fire Weather Index (FWI)” calculation, another is acquisition of sensor value and use of deep learning model for prediction. The last one is to map the temperature of the surface by using “Unmanned Aerial Vehicle (UAV)” [13]. Through this algorithm, the fire can be detected with a high accuracy level.

V. CONCLUSION AND RECOMMENDATIONS

As per the above discussion, it can be stated that, forest-monitoring process is a major area of concern across the world for the restoration of forests. After restoring forests with the help of preventing deforestation activities, carbon footprint can be reduced drastically. As per the findings of this research, AI-based tools and machine learning algorithms are effective to detect forest fire and predict forest cover changes with high accuracy. Some major challenges are found, in terms of the remote sensing process used within forest monitoring. Big data used in remote sensing are the main challenge in terms of computing, processing, representation, identification and many others.

However, it can be recommended that, using an advanced technical team, skilled persons with statistical knowledge will be effective for mitigating these big data challenges. The interview data revealed that, forest professionals think that lack of government interventions and skilled people are the barrier to use AI tools within forest monitoring systems. As per the interview data and secondary sources, it can be concluded that, use of machine learning algorithms and wireless network

within forest monitoring can be effective to increase the accuracy of the overall monitoring process and restore forest. As a result, it can be stated that, introducing government policies to grant sufficient financial resources to develop advanced technical teams and projects aimed at forest monitoring will be effective to enhance the accuracy of this process.

REFERENCES

- [1] Theglobalobservatory (2020). How Can Artificial Intelligence Help Curb Deforestation in the Amazon?. Retrieved on 24th March 2022 from: <https://theglobalobservatory.org/2020/11/how-can-artificial-intelligence-help-curb-deforestation-amazon/>
- [2] Weforum (2021). The space AI keeping tabs on COP26 carbon pledges. Retrieved on 24th March 2022 from: <https://www.weforum.org/agenda/2021/12/satellites-ai-saving-planet/>
- [3] Pratihast, A. K., DeVries, B., Avitabile, V., De Bruin, S., Herold, M., & Bergsma, A. (2016). Design and implementation of an interactive web-based near real-time forest monitoring system. *PloS one*, 11(3), e0150935. https://www.scienceopen.com/document_file/949d63d9-7108-45fb-bc86-22f010e3e5d4/PubMedCentral/949d63d9-7108-45fb-bc86-22f010e3e5d4.pdf
- [4] Kalantar, B., Ueda, N., Idrees, M.O., Janizadeh, S., Ahmadi, K. & Shabani, F. (2020). Forest fire susceptibility prediction based on machine learning models with resampling algorithms on remote sensing data. *Remote Sensing*, 12(22), 3682. https://mdpi-res.com/d_attachment/remotesensing/remotesensing-12-03682/article_deploy/remotesensing-12-03682-v2.pdf
- [5] Allmark, P., & Machaczek, K. (2018). Realism and Pragmatism in a mixed methods study. *Journal of advanced nursing*, 74(6), 1301-1309. <http://shura.shu.ac.uk/18205/1/Allmark-RealismAndPragmatism%28AM%29.pdf>
- [6] Al-Ababneh, M. M. (2020). Linking ontology, epistemology and research methodology. *Science & Philosophy*, 8(1), 75-91. <http://eir.is.it/ojs/index.php/scienceandphilosophy/article/viewFile/500732>
- [7] Woiceshyn, J., & Daellenbach, U. (2018). Evaluating inductive vs deductive research in management studies: Implications for authors, editors, and reviewers. *Qualitative Research in Organizations and Management: An International Journal*, 13(2), 183-195. https://www.researchgate.net/profile/Jaana-Woiceshyn-2/publication/324596659_Evaluating_Inductive_versus_Deductive_Research_in_Management_Studies_Implications_for_Authors_Editors_and_Reviewers/links/5b78d9aa92851c1e121f7477/Evaluating-Inductive-versus-Deductive-Research-in-Management-Studies-Implications-for-Authors-Editors-and-Reviewers.pdf
- [8] Olabode, S. O., Olateju, O. I., & Bakare, A. A. (2019). An assessment of the reliability of secondary data in management science research. *International Journal of Business and Management Review*, 7(3), 27-43. https://www.researchgate.net/profile/Akeem-Bakare-2/publication/344346438_AN_ASSESSMENT_OF_THE_RELIABILITY_OF_SECONDARY_DATA_IN_MANAGEMENT_SCIENCE_RESEARCH/links/5f6a7ff0a6fdcc0086346109/AN-ASSESSMENT-OF-THE-RELIABILITY-OF-SECONDARY-DATA-IN-MANAGEMENT-SCIENCE-RESEARCH
- [9] Johnston, M.P. (2017). Secondary data analysis: A method of which the time has come. *Qualitative and quantitative methods in libraries*, 3(3), 619-626. <http://www.qqml-journal.net/index.php/qqml/article/view/169/170>
- [10] Scharp, K. M., & Sanders, M. L. (2019). What is a theme? Teaching thematic analysis in qualitative communication research methods. *Communication Teacher*, 33(2), 117-121. https://www.researchgate.net/profile/Kristina-Scharp/publication/328675228_What_is_a_theme_Teaching_thematic_analysis_in_qualitative_communication_research_methods/links/5be360a392851c6b27aee2f6/What-is-a-theme-Teaching-thematic-analysis-in-qualitative-communication-research-methods.pdf
- [11] Chi, M., Plaza, A., Benediktsson, J. A., Sun, Z., Shen, J., & Zhu, Y. (2016). Big data for remote sensing: Challenges and opportunities.

Proceedings of the IEEE, 104(11), 2207-2219. <https://www.umbc.edu/rssipl/people/aplaza/Papers/Journals/2016.PIEEE.Bigdata.pdf>

- [12] Weforum (2021). These 3 start-ups are bringing cutting-edge tech to forest restoration. Retrieved on 24th March 2022 from: <https://www.weforum.org/agenda/2021/04/reforestation-start-ups-technology-trillion-trees/>
- [13] Benzekri, W., El Moussati, A., Moussaoui, O., & Berrajaa, M. (2020). 'Early forest fire detection system using wireless sensor network and deep learning. Int. J. Adv. Comput. Sci. Appl, 11(5), 496-503.